

KUAN-FU FENG

Postdoctoral Researcher

A geophysicist specializing in observational seismology with experience in applied science. Hands-on experience in geophysical experiments. Skilled in acquiring, processing, and analyzing time series data. Expertise in geoscience-related software and programming in Linux environments.

EXPERIENCE

Salt Lake City, UT Dec 2024 - Present	Postdoctoral Researcher <i>University of Utah, USA</i> - Conducting geophysical analysis to evaluate groundwater resource - Collaborating with interdisciplinary teams to improve hydro-geophysical monitoring methodologies
Seattle, WA Jun 2023 - Nov 2024	Postdoctoral Scholar <i>University of Washington, USA</i> - Engaged in developing data-driven models to quantify groundwater-surface interactions using seismic datasets - Implementing cloud computing technologies for seismic data processing - Contributing to NoisePy, an open-source software package dedicated to environmental seismology - Serving as an instructor for hands-on workshops at "Seismic Data Mining on the Cloud 101"
Salt Lake City, UT Jun 2022 - May 2023	Postdoctoral Scholar <i>University of Utah, USA</i> - Leveraging over 10+ TB of high-resolution time-series data for seismic analysis - Exploring ambient noise sources' effects across various scales of seismic arrays

EDUCATION

Sep 2017 - Jan 2022	PhD – Geosciences, National Taiwan University
Sep 2014 - Jun 2016	Master – Geosciences, National Taiwan University
Sep 2010 - Jan 2014	Bachelor – Earth and Environment Sciences, National Chung Cheng University

SKILLS

- Time series analysis, Signal Processing, Statistical analysis
- Programming: Bash Script, Fortran, C, Python, Scikit learn, Jupyter notebook
 - Dara visualization: ParaView, Adobe Illustrator
 - Cloud environment: AWS

PRE-PRINTS

- Feng, K.-F.***, M. Denolle, F-C. Lin and T. van Dam. (2024) A decadal survey of the near-surface seismic velocity response to hydrological variations in Utah, United States. <https://doi.org/10.31223/X52425> (under review in the *Journal of Geophysical Research: Solid Earth*)
- Denolle, M. A., Tape, C., Bozdag, E., Wang, Y., Waldhauser, F., Gabriel, A., Braunmiller, J., Chow, B., Ding, L., **Feng, K.-F.**, Ghosh, A., Groebner, N., Gupta, A., Krauss, Z., et al. (2024). Training the Next Generation of Seismologists: Delivering Research-Grade Software Education for Cloud and HPC Computing through Diverse Training Modalities. <https://doi.org/10.48550/arXiv.2409.19147> (submitted to *Seismological Research Letters*)

PUBLICATIONS

- Feng, K.-F.***, H.-H. Huang*, Ya-Ju Hsu and Y.-M. Wu (2021), Controls on seasonal variations of crustal seismic velocity in Taiwan using single-station cross-component analysis of ambient noise interferometry, *J. Geophys. Res., Solid Earth*, e2021JB022650. <https://doi.org/10.1029/2021JB022650>.
- Feng, K.-F.**, H.-H. Huang* and Y.-M. Wu (2020), Detecting pre-eruptive magmatic processes of the 2018 eruption at Kilauea, Hawaii volcano with ambient noise interferometry. *Earth, Planets and Space*, 72:74. <https://doi.org/10.1186/s40623-020-01199-x>

CONTACT INFORMATION

ADDRESS

Frederick Albert Sutton Building, Rm 365, 115 S 1460 E, Salt Lake City, UT 84112

EMAIL

kuanfufeng@gmail.com

LinkedIn PROFILE

<https://www.linkedin.com/in/kuan-fu-feng-48119b27a/>

ACHIEVEMENT

- Contributed to open-source seismic analysis tools
- Served as a hands-on workshop instructor
- Co-advised a PhD project

PROFESSIONAL SERVICES/ VOLUNTEER EXPERIENCES

- Nodal seismic array deployment
- Summer Student Lecture Convenor
- Journal peer-reviewer

LANGUAGES

Mandarin
Taiwanese
English